

Work Stream 3: Pre-Singularity Journey & Turbulence Modeling

MechanicaFluidorum Program & Community Contributors (esp. u/babainottawa)

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1 Introduction

As noted by u/babainottawa, while the singular limit itself lies outside the validity of the model that produces it, the *journey* of the singularity contains valuable information. The extreme vortex stretching observed in the regime where the continuum model still applies offers a unique mathematical testbed for turbulence models.

Which part of the journey is usable. Because the collapsing core keeps a radial Reynolds number of order one, its scales are diffusive, $\ell_r \simeq \sqrt{\nu t}$ and $u \simeq \sqrt{\nu/t}$ (up to factors $\tau^{\pm h}$, $h = 1/200$), and are essentially independent of the initial size of the vortex. In a liquid such as water the first constitutive limit is not compressibility but cavitation: the core pressure deficit $\sim \rho u^2$ reaches $p_\infty - p_v \approx 0.1$ MPa at $u \approx 14$ m/s, about 5 ns before blow-up, when $\ell_r \approx 70$ nm (with the exact deficit of a Gaussian core, $1.70 \rho u^2$, at $u \approx 7.6$ m/s, ≈ 17 ns, $\ell_r \approx 130$ nm; see Work Stream 4). Compressibility ($\text{Ma} = 0.3$) follows three decades later, at ≈ 5 –7 ps ($\ell_r \approx 3$ nm); in air the corresponding window closes at ≈ 1.5 –1.8 ns. The trajectory *before* these times—from the initial vortex down to core radii of tens of nanometres in water, and to the Mach limit in gases—is smooth, resolved, and physically meaningful. That is the segment we propose as a benchmark.

2 An Exact Two-Scale Closure

The OpenAI construction splits the velocity into a mean collapsing vortex u_B and oscillatory pulses w , confined to an annulus, whose averaged momentum flux $T = \langle w \otimes w \rangle$ is controlled explicitly:

$$\partial_t u_B + (u_B \cdot \nabla) u_B - \nu \Delta u_B + \nabla p_B = -\nabla \cdot T + f_{\text{smooth}}. \quad (1)$$

This is a Reynolds decomposition with an *exact* closure: energy flows from the mean shear into the pulses, and the averaged pulse flux drives the mean vortex into self-similar collapse. Three concrete uses follow.

1. **A priori test of subgrid-scale models.** Evaluate Smagorinsky, dynamic (Germano-identity), gradient/nonlinear, and backscatter-capable closures on u_B and compare the modelled stress with T in sign, anisotropy, and radial flux. Purely dissipative eddy-viscosity models cannot produce a stress whose work *concentrates* the mean flow; how, and how badly, they fail can be measured rather than argued.
2. **Code verification near singular behaviour.** The method of manufactured solutions is the standard tool for verifying CFD codes (Roache 2002). Truncations of the construction at finite τ within the usable window are smooth manufactured solutions with a known, very steep but controlled multiscale structure—a hard test of resolution, aliasing, and time stepping in DNS and LES solvers.

3. **Mean-flow/fluctuation feedback in swirling flows.** The pulse amplification relies on centrifugal and axial-shear instability criteria (Leibovich–Stewartson, Billant–Gallaire). The construction shows how such instabilities, arranged coherently, feed stress back into a concentrating vortex. This is of independent interest for vortex breakdown and tornado-like flows, and holds at moderate, physically realizable concentration, far from the singular limit.

The oscillating wave packets transfer energy to the core vortex in a highly structured, anisotropic manner; analysing this process provides a benchmark for dynamic models under transient, non-equilibrium conditions that no isotropic-turbulence dataset supplies.

3 Acknowledgments

We thank u/babainottawa for shifting the focus from the mathematical endpoint to the physical journey. *Rendons à César ce qui appartient à César.*